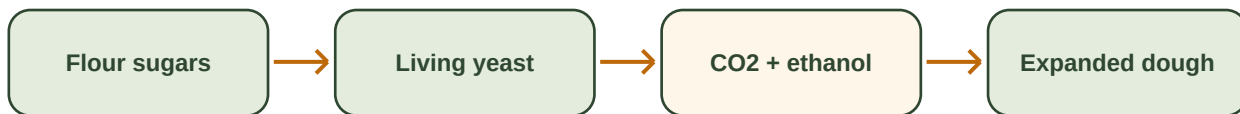


Kitchen Chemistry: Bread

A printable investigation of living yeast, fermentation, gas bubbles, gluten structure, ingredient ratios, temperature, and oven transformations.



Matter moves and changes form; the gas is trapped by the gluten network.

**The family
question**

**How can something too small to see fill a bowl with bubbles -
and why do those bubbles stay?**

Family name: _____

Date baked: _____ Flour used: _____

Print one workbook for the family. Younger learners can draw and dictate; middle learners can measure and explain; older learners can calculate and model.

1. Gather the family around the bowl

This is one investigation with different ways to contribute. Choose roles by readiness, not by rigid grade boundaries.

Younger learners

Touch flour and dough, notice bubbles and smells, draw the before-and-after loaf, and tell the change story aloud.

Middle learners

Use the scale, mark dough height, compare conditions, make a data table, and explain cause and effect.

Older learners

Calculate baker's percentages, model anaerobic fermentation, analyze variables, and explain heat-driven changes.

Materials

- [] 500 g flour
- [] 350 g lukewarm water
- [] 10 g salt
- [] 5-7 g yeast
- [] Mixing bowl and spoon
- [] Kitchen scale
- [] Towel or lid
- [] Loaf pan or baking sheet
- [] Marker or tape for dough height
- [] Optional thermometer

Adult safety checkpoint

An adult handles the hot oven, hot pan, and sharp tools. Check allergies before tasting. Lukewarm water should feel comfortable, not hot: excessive heat can kill yeast.

2. Measure a proportional system

Bread is not a random pile of ingredients. It is a system of ratios. Flour is the reference amount in baker's percentage.

Flour	500 g	100%	_____ g
Water	350 g	70%	_____ g
Salt	10 g	2%	_____ g
Yeast	5-7 g	1-1.4%	_____ g

Hydration calculation

$$\text{Water mass} / \text{flour mass} \times 100 = \text{hydration percentage}$$

Our calculation: _____

Prediction before mixing

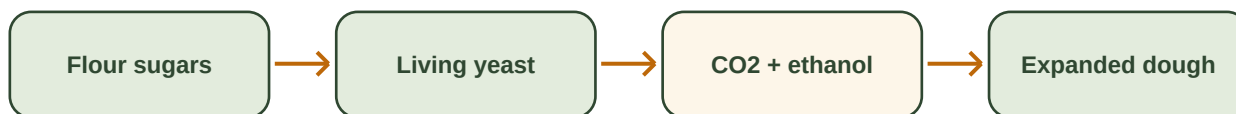
What will change after 60 minutes? Describe expected height, bubbles, smell, and texture.

Fair-test thinking

If you split the dough, choose one variable to change. Keep every other condition as similar as possible.

- ☐ Temperature
- ☐ Hydration
- ☐ Proofing time
- ☐ Flour type
- ☐ Kneading method

3. Watch living yeast change the dough



Matter moves and changes form; the gas is trapped by the gluten network.

Fermentation in plain language

Baker's yeast is a living fungus. Without enough oxygen, it can obtain energy from sugars through fermentation. Carbon dioxide gas and a small amount of ethanol are produced. Much of the ethanol evaporates during baking; the carbon dioxide inflates the dough.

Observation table

Time	Dough height	Bubbles / texture	Smell
Start			
20 min			
40 min			
60 min			

Evidence statement

I think fermentation occurred because I observed...

4. Find the invisible scaffold

Mixing and folding align flour proteins into a stretchy gluten network. Carbon dioxide does not make a loaf tall by itself: the dough also needs a structure that can stretch and hold the gas.

Windowpane test

Stretch a small piece of dough gently. Can it become thin enough for light to pass through before tearing? Draw or describe what you notice.

Temperature comparison

Condition	Starting temp	Rise after ____ min	What stayed the same?
Warmer sample			
Cooler sample			

Cause-and-effect claim

The evidence suggests temperature _____ yeast activity because _____.

5. Follow the loaf through the oven

The oven begins a second wave of transformations. Watch for evidence that heat is moving into the dough and changing matter.

- [] Gas expands and creates oven spring.
- [] Yeast activity stops as temperature rises beyond its survival range.
- [] Starch absorbs water and gelatinizes, helping set the crumb.
- [] Proteins firm and the loaf's structure becomes stable.
- [] Water evaporates from the surface.
- [] Browning reactions create new colors, aromas, and flavors in the crust.

Before and after

Shape: _____	Shape: _____
Surface: _____	Crust: _____
Inside: _____	Crumb: _____
Smell: _____	Smell: _____

Explain one oven change

6. Review the concepts together

Circle one answer for each question. Complete the interactive version online to have the registrar verify and seal the portfolio entry.

1. What is baker's yeast?

A. A living fungus B. A nonliving chemical powder C. A grain

2. Which gas expands the dough?

A. Oxygen B. Carbon dioxide C. Nitrogen

3. What does the gluten network do?

A. Traps gas B. Feeds yeast C. Produces heat

4. Why does warm dough usually rise faster?

A. Warmth speeds yeast activity B. Warmth adds gas C. Warmth makes flour

5. What does oven heat do?

A. Sets structure and browns crust B. Keeps yeast alive forever C. Removes gluten

6. How do you double a recipe?

A. Multiply every ingredient by two B. Double only flour C. Guess

Family explanation

Use the words yeast, sugar, carbon dioxide, gluten, ratio, temperature, and heat to explain how ingredients became bread.

7. Prepare the portfolio story

The registrar recognizes demonstrated understanding, careful observation, and a real artifact. Clock time is optional metadata; it is not the point of the lesson.

Concepts reviewed for credit

- ☐ Creation Science: yeast as a living organism and fermentation as a chemical process
- ☐ Creation Science: carbon dioxide production, gluten structure, and evidence of change
- ☐ Creation Science: temperature, heat transfer, starch/protein changes, and browning
- ☐ Applied Mathematics: mass, ratios, hydration percentage, and proportional scaling
- ☐ Scientific communication: observation, cause-and-effect explanation, and a next-test prediction

What did your family actually observe?

What single variable would you change next time, and what do you predict?

Portfolio evidence

- ☐ Photo of the finished loaf
- ☐ Photo of the crumb
- ☐ Observation table
- ☐ Calculation or scaled recipe
- ☐ One-sentence explanation from each learner

Finish online

Open Dear Adeline > Today > Kitchen Chemistry: Bread. Complete the six-question review, add your observations and next test, then select 'Review concepts & create portfolio entry.'